EL7 Worked Examples: Applications of Exponential and Logarithmic Functions

Recording companion for the EL7 CheckIt support video — Math 172, Santa Ana College

How to use this document

- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

Am I looking for the amount or for the time? If it is the time, the variable is in the exponent, so I need a logarithm. Write the model with the numbers in it first: the rate as a decimal, the right number of compounding periods, and 1 minus the rate when value is lost. To find an amount, plug in the time. To find the time, divide so the power is alone, switch to logarithmic form, and use natural logs on the calculator.

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Example 1: Set Up 1

$$A(t)=P(1+r/n)^{nt}$$

Question

A certificate of deposit starts with $15,000 and earns 4% interest, compounded quarterly. Which function gives the amount after t years?

- **A.** $A(t)=15000(1+4/4)^{4t}$
- **B.** $A(t)=15000(1+0.04/4)^{4t}$
- **C.** $A(t)=15000(1+0.04/4)^{t}$
- **D.** $A(t)=15000(1+0.04)^{4t}$

Answer and reasoning

The starting amount is $P=15000$. The rate 4% is $r=0.04$ as a decimal. Quarterly means 4 times a year, so $n=4$, and n appears twice: once dividing the rate and once multiplying t. Inside the parentheses, 0.04 divided by 4 is 0.01, so the base is 1.01. Using 4 instead of 0.04 would double the money every quarter. **The model is $A(t)=15000(1.01)^{4t}$, written with the rate 0.04 divided by 4.**

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Example 2: Set Up 2

$$A(t)=A(1-r)^t$$

Question

A car is worth \$40,000 and loses 5% of its value each year. Which function gives its value after t years?

- **A.** $A(t)=40000(1+0.05)^t$
- **B.** $A(t)=40000(0.05)^t$
- **C.** $A(t)=40000(0.95)^t$
- **D.** $A(t)=40000(1-5)^t$

Answer and reasoning

Losing 5% each year means the car keeps 95% of its value each year: $1-0.05=0.95$. Using $1+0.05$ would make the value grow. Using 0.05 alone would keep only 5% each year. **The model is $A(t)=40000(0.95)^t$, which is $40000(1-0.05)^t$.**

Example 3: Set Up 3

$$A(t)=A(1/2)^{t/h}$$

Question

Cesium-137 has a half-life of 30 years. A sample starts with 400 micrograms. Which function gives the amount left after t years?

- **A.** $A(t)=400(1/2)^{t/30}$
- **B.** $A(t)=400(1/2)^{30t}$
- **C.** $A(t)=400(1/2)^{30/t}$
- **D.** $A(t)=400(2)^{t/30}$

Answer and reasoning

The exponent counts how many half-lives have passed: time divided by the half-life, $t/30$. After 30 years the exponent is 1, so half is left. Check with numbers: at $t=30$, $400((2)^{1)/200}$. A base of 2 would double the sample instead. **The model is $\boldsymbol{A(t)=400(1/2)^{t/30}}$.**

Example 4: Set Up 4

$$P(t)=a\cdot b^t$$

Question

A colony starts with 2000 bacteria. After 3 days there are 6750. What is b in $P(t)=2000\cdot b^t$?

- **A.** $b=3.375$
- **B.** $b=1.125$
- **C.** about 1583
- **D.** $b=1.5$

Answer and reasoning

Plug in the second data point: $6750=2000\cdot b^3$. Divide by 2000: $b^3=3.375$. Take the cube root: $b=1.5$. Check: $2000\cdot 1.5^3=2000\cdot 3.375=6750$. The value 3.375 is b^3, not b, and dividing 3.375 by 3 is not a cube root. About 1583 is the average number added per day, which describes a line, not an exponential. **The growth factor is $b=1.5$.**

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Example 5: Amount 1

$$A(t)=15000(1.01)^{4t}$$

Question

How much will be in the certificate of deposit after 10 years? Round to the nearest dollar.

A. $22,204
B. $16,569
C. $22,333
D. $72,015

Answer and reasoning

Plug in $t=10$: the exponent is $4\cdot 10=40$, and the base is 1.01, because 0.04 divided by 4 is 0.01. So $A(10)=15000(1.01)^{40}\approx 22332.96$. The answer $22,204 compounds once a year instead of quarterly, $16,569 uses an exponent of 10 instead of 40, and $72,015 forgets to divide the rate by 4. **The account will have about $22,333.**

Example 6: Amount 2

$$A(t)=40000(0.95)^t$$

Question

What will the car be worth after 6 years? Round to the nearest dollar.

A. $29,404
B. $28,000
C. $53,604
D. $38,000

Answer and reasoning

Plug in $t=6$: $A(6)=40000(0.95)^6 \approx 29403.68$. The answer $28,000 takes off $2,000 every year, which is a line, but each year the car loses 5% of a smaller value. The answer $53,604 uses 1.05, so the car gains value, and $38,000 applies the 5% loss only once. **The car will be worth about $29,404.**

Example 7: Amount 3

$$A(t)=400(1/2)^{t/30}$$

Question

How much cesium-137 will be left after 90 years?

A. about 133 micrograms
B. 50 micrograms
C. about 0 micrograms
D. 3200 micrograms

Answer and reasoning

The exponent is $90/30=3$, so three half-lives have passed: $400\cdot((2)^{3)/(400/)\{8\}=50}$. In words: 400, then 200, then 100, then 50. Dividing by 3 gives about 133, which is not how halving works. **There will be 50 micrograms left.**

Example 8: Amount 4

$$P(t)=2000\cdot 1.5^{t}$$

Question

How many bacteria will there be after 5 days? Round to the nearest whole number.

- **A.** 15,000
- **B.** 9,917
- **C.** 875,788
- **D.** 15,188

Answer and reasoning

Plug in $t=5$: $2000\cdot 1.5^{5}=2000\cdot 7.59375=15187.5$, which rounds to 15,188. The answer 15,000 multiplies 1.5 by 5 instead of raising it to the 5th power. The answer 875,788 uses 3.375 as the growth factor. **There will be about 15,188 bacteria.**

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Example 9: Time 1

$$28590=20000(1.015)^{2t}$$

Question

A savings account starts with $20,000 and earns 3% compounded semiannually. How many years until it has $28,590?

A. about 12 years
B. about 24 years
C. about 6 years
D. about 0.7 years

Answer and reasoning

Divide by 20000: $1.015^{2t}=1.4295$. Switch forms: $2t=\log_{1.015}(1.4295)=(\ln 1.4295)/(\ln 1.015)\approx 24.0$. Then divide by 2: $t\approx 12$. The answer 24 is the number of half-year periods, not years. **It takes 12 years.**

Example 10: Time 2

$$36069=50000(0.96)^t$$

Question

A boat worth \$50,000 loses 4% of its value each year. After how many years is it worth \$36,069?

A. about 7 years
B. about 0.75 years
C. about 8 years
D. about negative 8.3 years

Answer and reasoning

Divide by 50000: $0.96^t=0.72138$. Switch forms: $t=(\ln 0.72138)/(\ln 0.96)\approx 8.0$. Both logarithms are negative, so the answer is positive. A negative time means the model used 1.04, which is growth. About 7 years comes from taking off \$2,000 every year, which is a line. **It takes 8 years.**

Example 11: Time 3

$$25=200(1/2)^{t/5730}$$

Question

Carbon-14 has a half-life of 5730 years. A sample starts with 200 grams. After how long will 25 grams be left?

A. 3 years
B. 22,920 years
C. 45,840 years
D. 17,190 years

Answer and reasoning

Divide by 200: $((2)^{t/5730)/(1/)\{8\}}$. Since $(8=((2)^{3)/)/}$, the exponent is 3: $t/5730=3$, so $t=17190$. The answer 3 is the number of half-lives, not years. 45,840 years comes from 8 half-lives, mixing up "one eighth left" with "8 halvings". **It takes 17,190 years.**

Example 12: Time 4

$$34172 = 2000 \cdot 1.5^{t}$$

Question

For the bacteria colony, after how many days will there be 34,172 bacteria?

A. about 11.4 days
B. about 7 days
C. about 25.7 days
D. about 20.3 days

Answer and reasoning

Divide by 2000: $1.5^{t} = 17.086$. Switch forms: $t = (\ln 17.086)/(\ln 1.5) \approx 7.0$. About 11.4 divides 17.086 by 1.5 instead of using a logarithm. About 25.7 skips dividing by 2000. **It takes 7 days.**

Example 13: Slip 1

$$A(t)=8000(1+4.5)^t$$

Question

A student wrote this model for \$8,000 earning 4.5% interest, compounded annually. What is the mistake, if any?

A. The exponent should be 4.5 t, because the rate goes in the exponent.
B. The rate should be the decimal 0.045, not 4.5.
C. The plus sign should be a minus sign, because interest is taken out.
D. There is no mistake.

Answer and reasoning

A percent has to become a decimal before it goes in the model: 4.5% is *0.045*. With 4.5, the base is 5.5, so the money would multiply by 5.5 every year. **The correct model is $A(t)=8000(1.045)^t$.**

Example 14: Slip 2

$$A(t)=P(1+0.03/6)^{6t}$$

Question

A student wrote this model for an account earning 3% interest, compounded semiannually. What is the mistake, if any?

A. The rate should be 3, because the percent is already given.
B. The 6 should appear only in the exponent, not under the rate.
C. Semiannually means twice a year, so both 6s should be 2.
D. There is no mistake.

Answer and reasoning

Semiannually means every six months, which is 2 times a year. The 6 counts months, not compounding periods. Both places need $n=2$: the rate is 0.03 divided by 2, which is 0.015. **The correct model is $A(t)=P(1.015)^{2t}$.**

Example 15: Slip 3

$$A(t)=30000(1.06)^t$$

Question

A student wrote this model for a \$30,000 RV that loses 6% of its value each year. What is the mistake, if any?

A. Losing value means the factor should be 0.94.
B. The factor should be 0.06, the amount lost each year.
C. The exponent should be 6 t, to match the 6 percent.
D. There is no mistake.

Answer and reasoning

Losing 6% each year means keeping 94%: $1-0.06=0.94$. The student's 1.06 makes the RV gain 6% a year. **The correct model is $A(t)=30000(0.94)^t$.**

Example 16: Slip 4

$$5000(1.2)^t=20000$$
$$6000^t=20000$$

Question

A student wrote these lines. What is the mistake, if any?

A. There is no mistake.
B. The student should divide 20000 by 1.2 to undo the power.
C. The student should subtract 5000 from both sides first.
D. Only 1.2 is raised to the power t. Divide by 5000 first.

Answer and reasoning

The exponent belongs to 1.2 only. Divide by 5000 first: $1.2^t=4$. Then $t=(\ln 4)/(\ln 1.2)\approx 7.6$. The student's line gives about 1.14, which is wrong. **The correct next line is $1.2^t=4$.**

Example 17: Slip 5

$$A(t)=80(1/2)^{t/30}$$
$$A(60)=80(1/2)^{2}=20$$

Question

A student wrote these lines for a sample of 80 micrograms with a half-life of 30 years. What is the mistake, if any?

A. The exponent should be 30 t.
B. After 60 years, only one half-life has passed.
C. There is no mistake.
D. The base should be 2.

Answer and reasoning

The exponent is time over half-life: $60/30=2$, so two half-lives have passed. Halving twice: 80, then 40, then 20. The student's work is correct. **There is no mistake: 20 micrograms are left.**